COURSE: TEXT ANALYTICS

TEXT PREPROCESSING PIPELINE FOR SOCIAL MEDIA DATA

Portfolio Project

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CONTEXT & BACKGROUND

Social media data is highly unstructured and complex in nature. It contains multiple forms of noise, such as:

- Emojis and special characters (e.g., ,)
- Hashtags (e.g., #trending)
- User mentions (e.g., @user)
- URLs and hyperlinks
- Slang and abbreviations (e.g., “LOL”, “brb”)
- Typographical errors and inconsistent language

This inherent noise makes raw data unsuitable for direct analysis. Therefore, text preprocessing is a critical step to clean, standardize, and transform the data into a structured format suitable for meaningful analysis and insights.

OBJECTIVE OF THE STUDY

- To clean and prepare social media text data
- To remove unwanted noise and irrelevant content
- To transform raw text into structured format
- To build an effective preprocessing pipeline

DATASET OVERVIEW &
INITIAL ANALYSIS

<u>DATASET:</u>
Sample Twitter-based text dataset

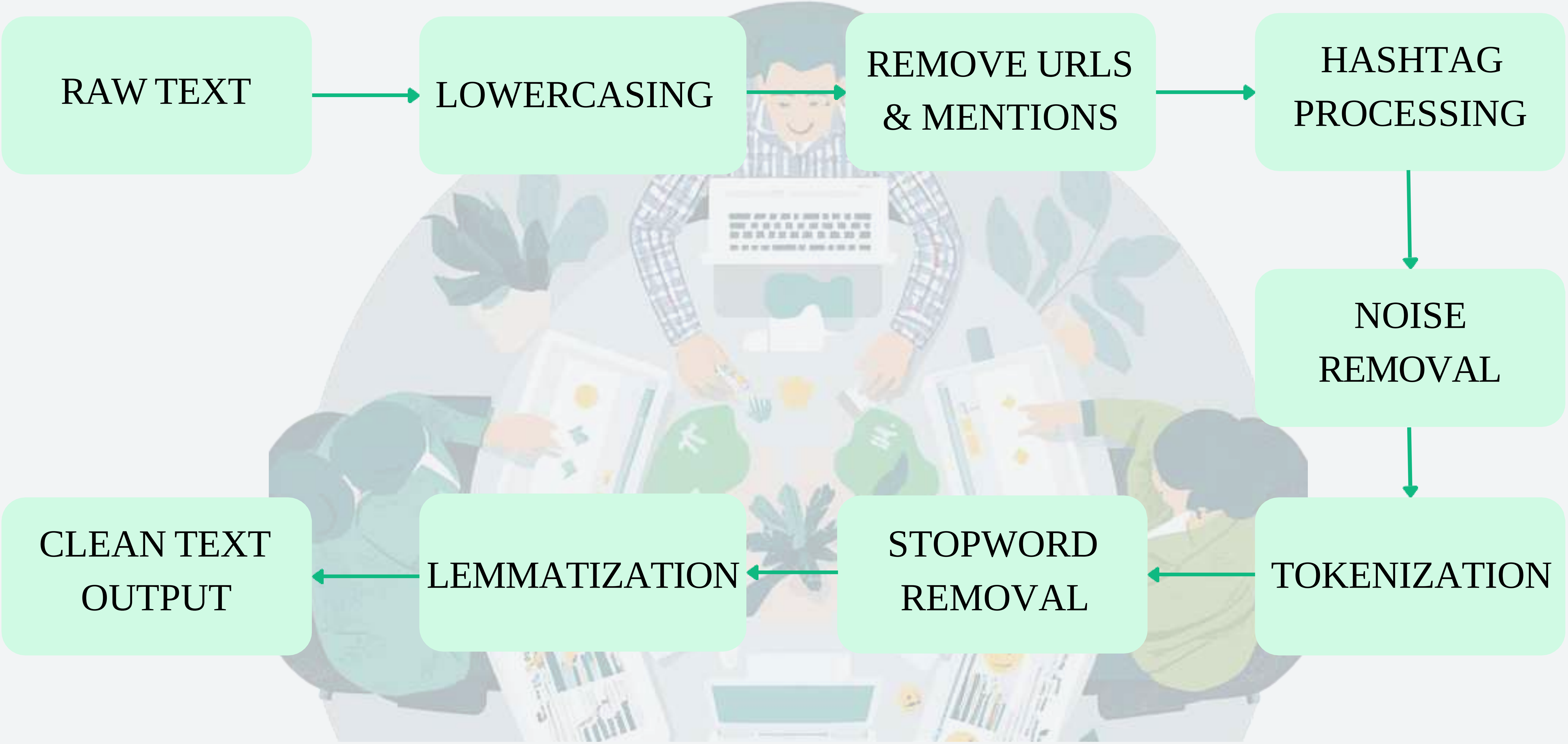
<u>NATURE OF DATA:</u>

- Informal language and abbreviations
- Presence of special characters and symbols
- High variability in text length

<u>PRELIMINARY OBSERVATIONS:</u>

- Significant noise in raw text
- Frequent use of non-standard expressions
- Need for systematic cleaning

TEXT PREPROCESSING FRAMEWORK



PREPROCESSING TECHNIQUES

Lowercasing

- Converts all text into lowercase
- Example: “Amazing Product” → “amazing product”
- Helps maintain uniformity and avoid duplication

Tokenization

- Breaks text into individual words (tokens)
- Example: “I love this product” → [“I”, “love”, “this”, “product”]
- Essential for further text processing

Stopword Removal

- Removes commonly used words (e.g., is, the, and)
- These words do not carry significant meaning
- Helps focus on important keywords

Noise Removal

- Removes unwanted elements like: URLs (http://...), Mentions (@user), Special characters & emojis
- Improves text clarity and quality

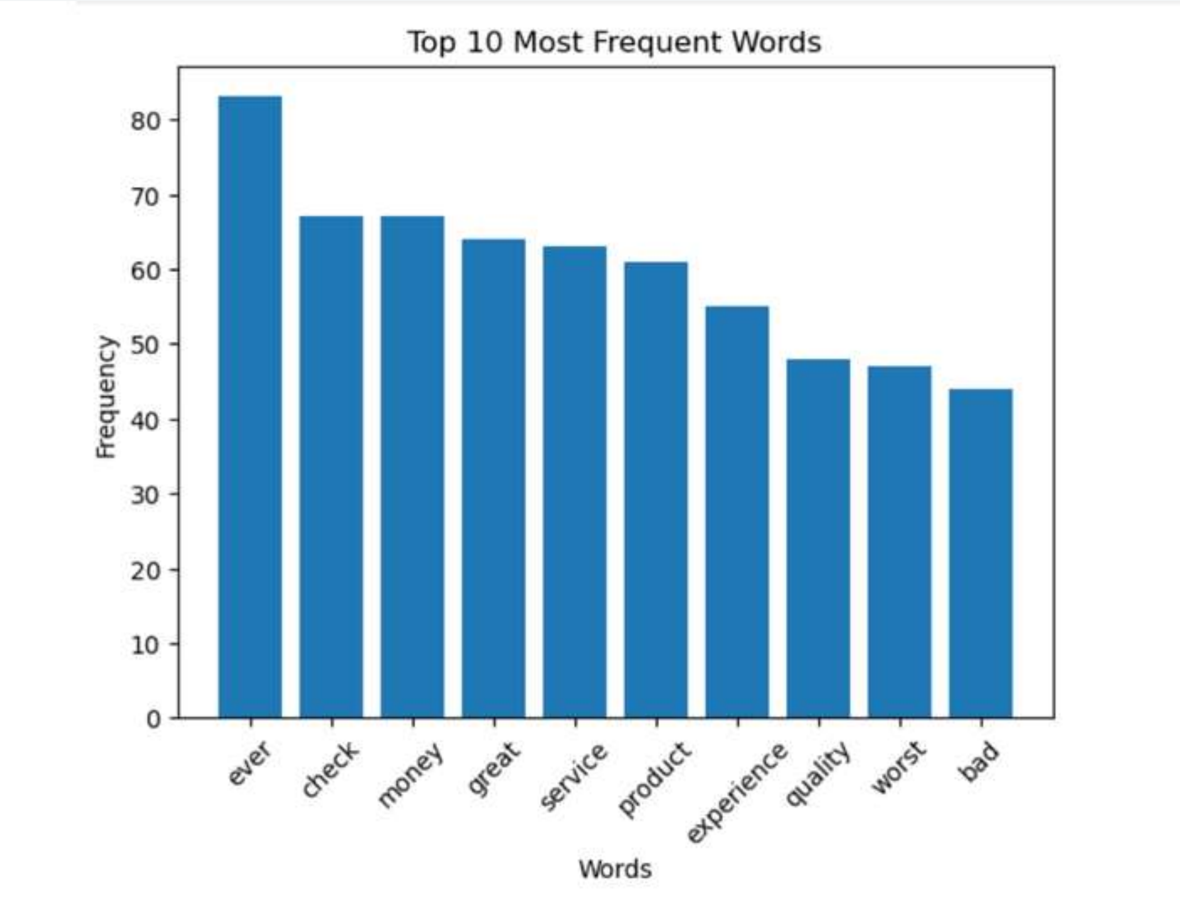
Hashtag Processing

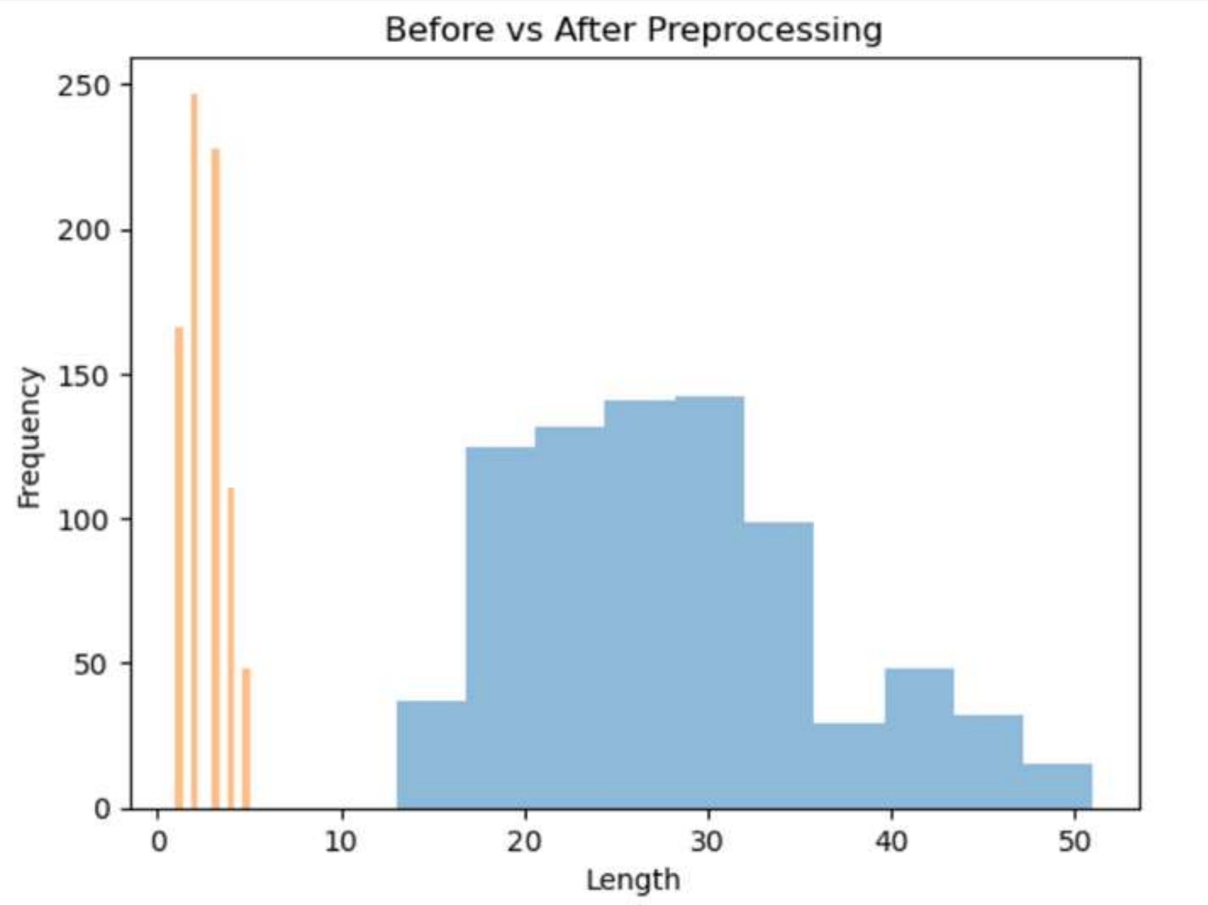
- Removes “#” but keeps meaningful word
- Example: #happy → happy
- Helps retain important context

Lemmatization

- Converts words to their base form
- Example: “running”, “runs” → “run”
- Produces meaningful and readable words (better than stemming)

RESULTS & VISUALIZATION



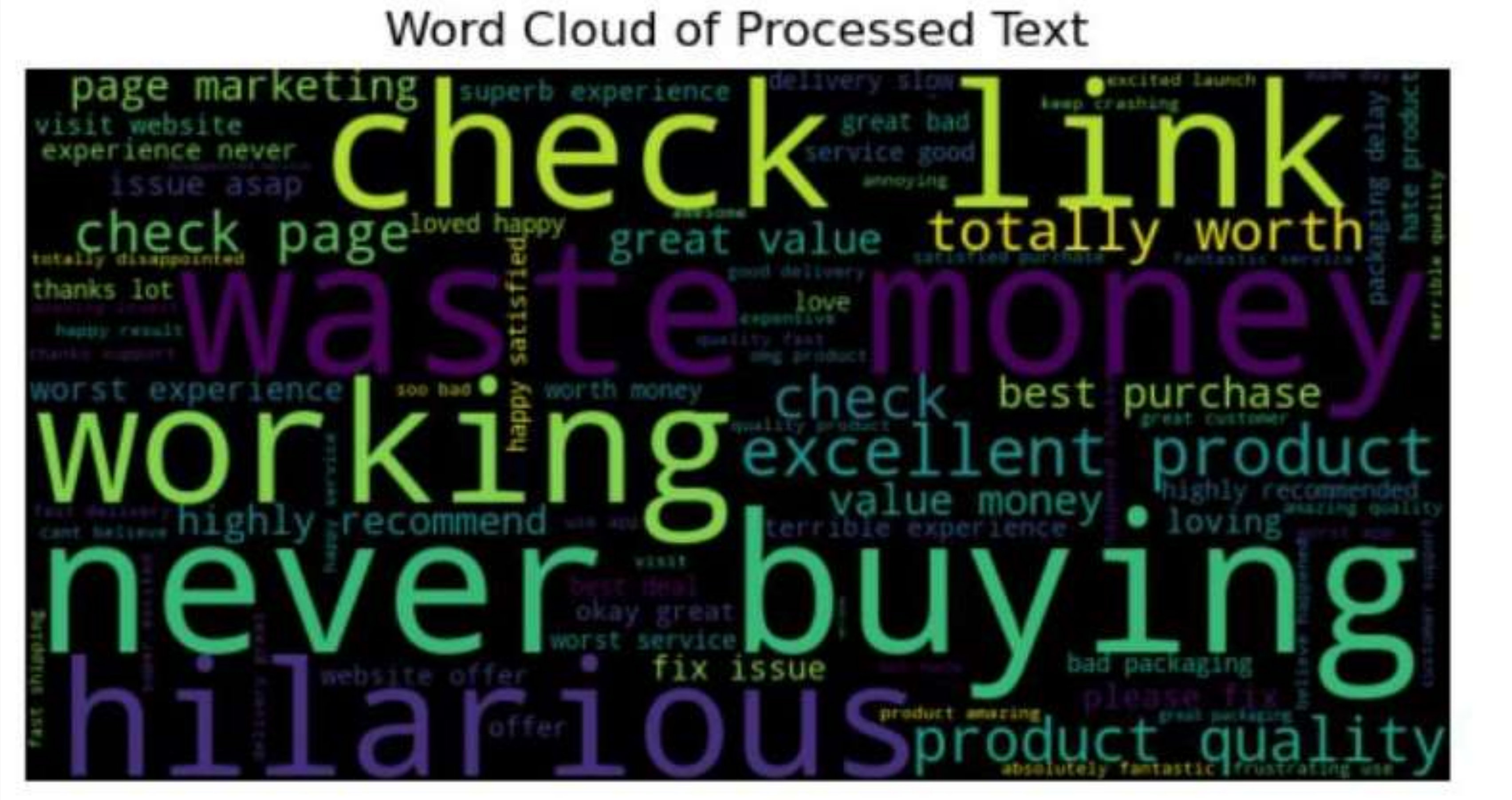


Visualization Insights:

- Word cloud highlights dominant themes in the dataset
- Bar chart shows most commonly used words
- Preprocessing reduced text complexity and noise

Results:

- Frequent words such as "product", "good", "service" were identified
- Noise like URLs, symbols, and emojis was removed
- Text became cleaner and more structured
- Stopwords and redundant words were successfully eliminated, improving the quality of the dataset



INSIGHTS, LIMITATIONS & CONCLUSION

KEY INSIGHTS:

- Social media data is highly noisy
- Preprocessing improves data quality
- Meaningful patterns can be extracted after cleaning

LIMITATIONS:

- Emojis are removed (not interpreted)
- Slang may not be fully handled
- Context like sarcasm is lost

CONCLUSION:

- Text preprocessing is essential for analyzing social media data effectively
- The pipeline transforms raw text into a clean and structured format
- It provides a strong foundation for further analysis like sentiment analysis and machine learning

THANK YOU!